

SYNTHESIS AND CHARACTERIZATION OF ZnO/g-C₃N₄ COMPOSITES FOR PHOTOCATALYTIC APPLICATIONS

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Low-temperature formation of carbon nitride from supramolecular precursors provides a model system for studying kinetically controlled structure formation and defect organization. In this work, ZnO/g-C₃N₄ composite materials were synthesized via thermal treatment of mechanically mixed precursors (melamine–thiobarbiturate supramolecular complexes[1] and zinc hydroxide) in an inert atmosphere using a low-temperature synthesis approach for g-C₃N₄[2].

Differential scanning calorimetry (DSC) analysis was performed to determine the optimal synthesis temperature. The results showed that the optimal synthesis temperature for the composite is 316 °C, which is higher than that of pure g-C₃N₄ (305 °C), indicating modification of the condensation pathway and effective energy landscape of condensation due to ZnO incorporation. The supramolecular structuring of the precursor system enables a preorganized condensation pathway leading to reduced structural randomness and non-random defect distribution, allowing a reduction of the synthesis temperature by approximately 200 °C compared to conventional g-C₃N₄ synthesis. The formation of the composite structure was confirmed by X-ray diffraction (XRD) and Fourier-transform infrared spectroscopy (FTIR), demonstrating the coexistence of both ZnO and g-C₃N₄ phases without formation of secondary phases.

Morphological analysis using scanning electron microscopy (SEM) revealed that at low ZnO content, ZnO particles are uniformly distributed on the layered g-C₃N₄ surface, forming a developed interface. With increasing ZnO content, particle agglomeration becomes dominant, leading to structural changes in the composite. Energy-dispersive X-ray spectroscopy (EDX) confirmed the presence of all key elements, including sulfur, indicating successful doping and preservation of the modified structure.

Optical properties were investigated using UV–Vis spectroscopy and Tauc plots. The obtained bandgap values lie between those of pure ZnO and g-C₃N₄.

Electron paramagnetic resonance (EPR) analysis revealed the presence of stable defect states. In addition to the characteristic signal of g-C₃N₄ ($g \approx 2.003$), an additional signal at $g \approx 2.01$ was detected, associated with sulfur-related defect centers stabilized at the ZnO/g-C₃N₄ interface.

It is shown that ZnO particles can serve as growth centers in g-C₃N₄ crystals, influencing defect formation and structural organization. The results indicate that the formation of ZnO/g-C₃N₄ composites is governed by supramolecular preorganization and kinetically controlled condensation, leading to structural inheritance and stabilization of sulfur-related defect centers at the heterointerface.

The work was supported by the Russian Science Foundation (project № 24-13-00355).